Information Retrieval Indexing

Lecture 2: Posting Lists, Compression & ANN Indexing

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Information Retrieval

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Outline

1. The Information Retrieval Process
2. Inverted Index Posting Lists
3. Query Processing Algorithms
4. Index Compression
5. Approximate k-NN Indexing for Dense Retrieval
6. Summary

Course Overview

1. Introduction to IR: VSM & Boolean Retrieval ✓
2. **Indexing: Sparse and Dense** ← Today
3. Query Understanding & Expansion
4. Evaluation of IR Systems
5. Ranking: BM25, Language Models, and Beyond
6. Embeddings & Neural Ranking
7. Dense Retrieval
8. Learning to Rank

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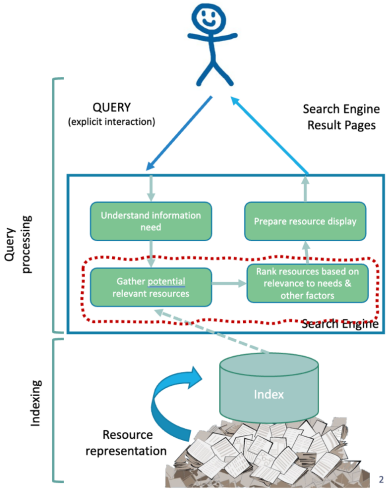
The Complete IR Pipeline



Text Preprocessing & Indexing

The Foundation of Search

Building an IR system happens in three main stages:

- **Stage 1: Offline Indexing**
  - Documents are preprocessed (tokenized, stemmed, etc.)
  - Inverted indexes are built and compressed
- **Stage 2: Online Query Processing**
  - Query is preprocessed **identically** to documents
  - Match terms in the index
- **Stage 3: Ranking**
  - Scored according to a model (TF-IDF, BM25, etc.)

Step-by-Step: Query Processing

Applying consistent preprocessing

Example

User Query: "Running Dogs in the Park"

1. **Tokenization:** [Running, Dogs, in, the, Park]
2. **Lowercasing:** [running, dogs, in, the, park]
3. **Stop word removal:** [running, dogs, park]
4. **Stemming:** [run, dog, park]

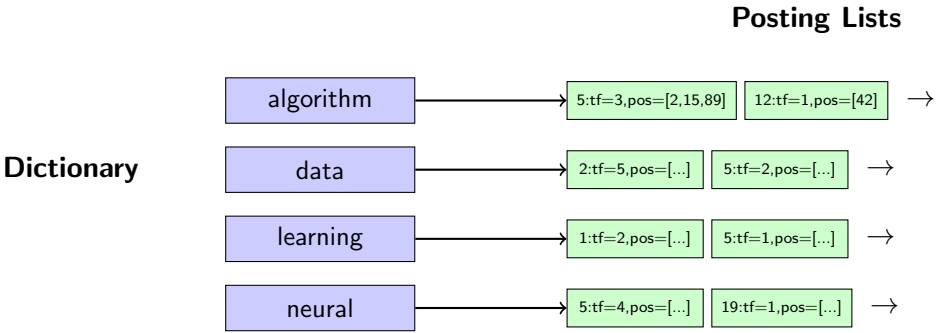
Processed Query: [run, dog, park]

The Consistency Principle

Preprocessing must be identical for indexing and retrieval!

Inverted Index: Quick Recap

Core data structure for text search



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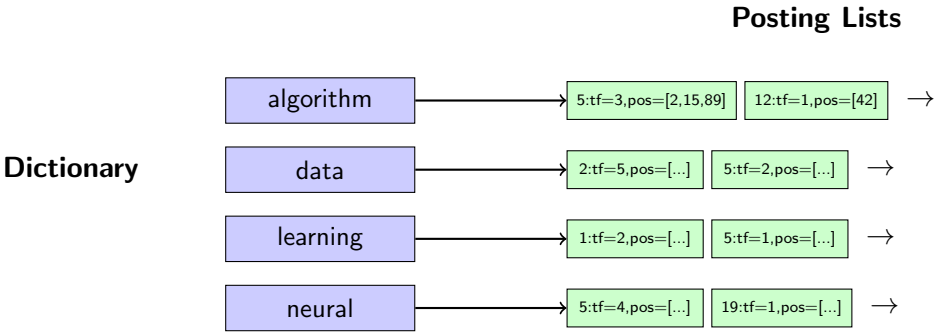

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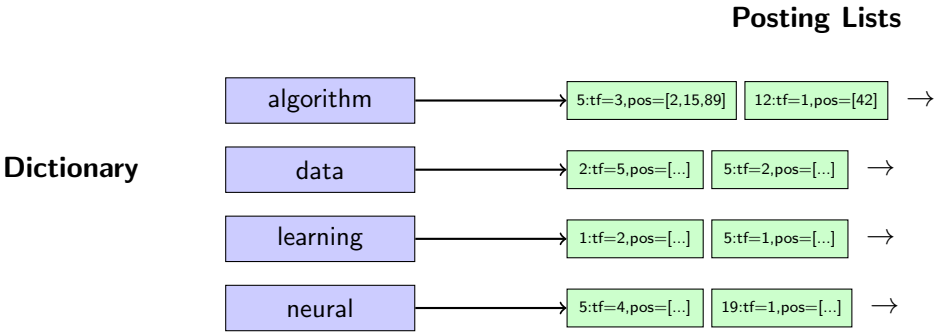

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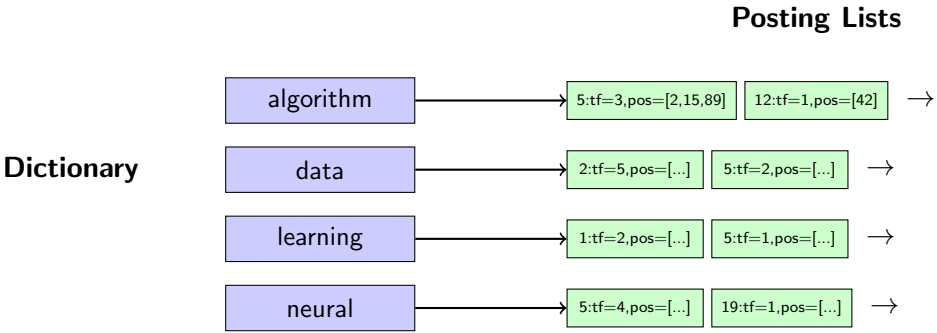

Each posting contains:

- Document ID

Posting List Formats (1/2)

Basic Format: DocID Only

$$\langle \text{docID}_1, \text{docID}_2, \text{docID}_3, \ldots \rangle$$

- Minimal information
- Boolean queries only
- Example: [5, 12, 27, 89, 102]

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- Example: [5, 12, 27, 89, 102]

With Frequencies

$$\langle (\text{docID}_1, \text{tf}_1), (\text{docID}_2, \text{tf}_2), \ldots \rangle$$

- Enables TF-IDF, BM25 scoring
- Example: [(5, 3), (12, 1), (27, 2), (89, 1), (102, 5)]

Posting List Formats (2/2)

With Positions (Full Inverted Index)

$$\langle(\text{docID}_1, \text{tf}_1, [\text{pos}_{1,1}, \text{pos}_{1,2}, \ldots]), \ldots\rangle$$

- Enables phrase queries, proximity scoring
- Example: [(5, 3, [2, 15, 89]), (12, 1, [42]), ...]
- Larger storage, but most flexible

Recommendation: Use positions for production systems (enables phrase queries)

Posting List Properties

Key Properties

1. **Sorted by DocID**
   - Enables efficient merge operations
   - Required for compression (gap encoding)
   - Allows skip pointers

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   - Common terms: long lists (e.g., "the": millions of docs)
   - Power-law distribution (Zipf's law)

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   - Common terms: long lists (e.g., "the": millions of docs)
   - Power-law distribution (Zipf's law)
3. **Stored Sequentially**
   - Good cache locality
   - Sequential disk reads (fast)
   - Compressed for storage efficiency

Skip Lists: Motivation

Problem

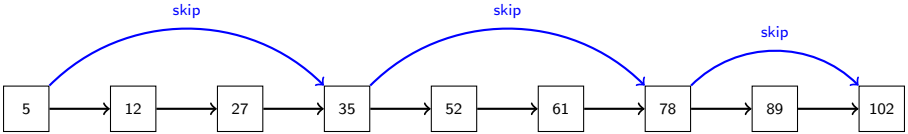
Merging long posting lists can be slow

- Query: "algorithm AND learning"
- "algorithm": 1M documents
- "learning": 500K documents
- Naive merge: scan both lists completely

Example: Merging Without Skip Lists

List A: [1, 5, 10, 15, 20, 25, 30, ...] (1M entries)
List B: [3, 6, 9, 12, 15, 18, 21, ...] (500K entries)
Finding intersection requires comparing **every** element:
Compare 1 vs 3, 5 vs 3, 5 vs 6, 10 vs 6, 10 vs 9, ...

Skip Lists: Structure



Posting List with Skip Pointers

Skip pointers properties:

- Placed at regular intervals (e.g., every $\sqrt{n}$ elements)
- Point to future positions in the list
- Store the maximum docID they skip over

Trade-off:

- ✓ Faster intersection (skip irrelevant parts)
- ✗ Extra storage for pointers

Skip Lists: Intersection Algorithm

```
Input: Two posting lists p_1, p_2 with skip pointers
Output: Intersection of p_1 and p_2

i_1 ← 0, i_2 ← 0 {Pointers to current positions}
result ← []
while i_1 < |p_1| AND i_2 < |p_2| do
  if p_1[i_1].docID = p_2[i_2].docID then
    Append p_1[i_1].docID to result
    i_1 ← i_1 + 1, i_2 ← i_2 + 1
  else if p_1[i_1].docID < p_2[i_2].docID then
    if p_1[i_1] has skip AND p_1[i_1].skip.maxDocID ≤ p_2[i_2].docID then
      i_1 ← p_1[i_1].skip {Use skip pointer}
    else
      i_1 ← i_1 + 1 {Advance normally}
    end if
  else
    if p_2[i_2] has skip AND p_2[i_2].skip.maxDocID ≤ p_1[i_1].docID then
      i_2 ← p_2[i_2].skip
    else
      i_2 ← i_2 + 1
    end if
  end if
end while
return result
```

Skip Lists: Example Intersection

Query: Find intersection of two lists

List A: [5, 12, 27, 35, 52, 61, 78, 89, 102] (skip every 3)
List B: [12, 35, 52, 89, 102, 150]

Trace:

1. Compare 5 vs 12 → 5 ¡ 12, skip from 5 to 35
2. Compare 35 vs 12 → 35 ¿ 12, advance B to 35
3. Compare 35 vs 35 → **Match!** → Output 35
4. Compare 52 vs 52 → **Match!** → Output 52
5. Compare 61 vs 89 → 61 ¡ 89, skip from 61 to 89
6. Compare 89 vs 89 → **Match!** → Output 89
7. Compare 102 vs 102 → **Match!** → Output 102
8. List A exhausted

Result: [35, 52, 89, 102]

Skip List Placement Strategy

How many skip pointers?

Trade-off between space and speedup

Common strategies:

1. $\sqrt{n}$ **spacing** (optimal for random access)
   - For list of length n, place $\sqrt{n}$ skip pointers
   - Each skip jumps $\sqrt{n}$ positions
   - Example: 10,000 elements $\rightarrow$ 100 skips, each spanning 100 elements
2. **Fixed interval** (simpler)
   - Skip every k elements (e.g., $k = 10$)
   - Easier to implement and maintain
3. **Multi-level skip lists**
   - Skip pointers at multiple granularities
   - Level 1: every 10, Level 2: every 100, etc.
   - More complex but faster for very long lists

Query Processing: The Task

Input

- Query: set of terms $q = \{t_1, t_2, \ldots, t_k\}$
- Inverted index with posting lists
- Scoring function (e.g., BM25, cosine similarity)

Output

- Ranked list of top-N documents by score
- Typically $N = 10$ or $N = 1000$

Key challenge: Efficiency

- Can't score all documents (millions/billions)

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Key challenge: Efficiency

- Can't score all documents (millions/billions)
- Must process posting lists cleverly
- Different algorithms for different scenarios

Two Main Paradigms

Document-at-a-Time (DAAT)

- Process one document completely
- Compute full score for that doc
- Move to next document
- Keep top-k heap

Analogy: Grading exams

- Grade all questions for student 1
- Then all questions for student 2
- Etc.

Term-at-a-Time (TAAT)

- Process one query term completely
- Update partial scores for all docs
- Move to next query term
- Accumulate scores

Analogy: Grading exams

- Grade question 1 for all students
- Then question 2 for all students
- Etc.

Which is better?

Depends on: query length, list lengths, scoring function, hardware

Document-at-a-Time (DAAT)

```
Input: Query terms {t_1, t_2, ..., t_k}, posting lists
Output: Top-N documents

Fetch posting lists for all query terms
heap ← empty min-heap of size N {Store top-N docs}
nextDocID ← minimum docID across all posting lists
while posting lists not exhausted do
    d ← nextDocID
    score ← 0
    for each term t_i in query do
        if t_i's posting list contains d then
            score ← score + termScore(t_i, d) {BM25, TF-IDF, etc.}
            Advance t_i's pointer
        end if
    end for
```

DAAT Example

Query: "neural learning"
Posting lists:

- neural: [5, 12, 27, 89]
- learning: [5, 27, 35, 89]

Processing (top-2):

1. **Doc 5:** Both terms present $\rightarrow$ score $=$ $\text{BM25}(\text{neural}, 5) + \text{BM25}(\text{learning}, 5) = 2.3$
   - Heap: [(5, 2.3)]
2. **Doc 12:** Only "neural" $\rightarrow$ score $= \text{BM25}(\text{neural}, 12) = 1.1$
   - Heap: [(5, 2.3), (12, 1.1)]
3. **Doc 27:** Both terms $\rightarrow$ score $= 3.1$
   - Heap: [(5, 2.3), (27, 3.1)] (evict doc 12)
4. **Doc 35:** Only "learning" $\rightarrow$ score $= 0.9$
   - Heap unchanged (score too low)
5. **Doc 89:** Both terms $\rightarrow$ score $= 2.5$

DAAT with Skip Pointers

Optimization: Use skip pointers to jump to next candidate document

```
Find document d that appears in at least one posting list
if d appears in all posting lists then
    Compute full score for d
else if d appears in some but not all lists then
    Compute partial score for d (only from matching terms)
end if
Use skip pointers to advance to next candidate efficiently
```

Key insight:

- Shortest posting list determines candidates
- Use skips to align longer lists to candidates
- Much faster when lists have very different lengths

Term-at-a-Time (TAAT)

```
Input: Query terms {t_1, t_2, ..., t_k}, posting lists
Output: Top-N documents

scores ← empty hash table (docID → score)
for each query term t_i do
    Fetch posting list for t_i
    for each document d in posting list do
        if d not in scores then
            scores[d] ← 0
        end if
        scores[d] ← scores[d] + termScore(t_i, d)
    end for
end for
Sort documents by score (descending)
return top-N documents
```

TAAT Example

Query: "neural learning"
Posting lists:

- neural: [5, 12, 27, 89]
- learning: [5, 27, 35, 89]

Processing:
Step 1: Process "neural"

- Doc 5: scores[5] $= 1.5$
- Doc 12: scores[12] $= 1.1$
- Doc 27: scores[27] $= 1.6$
- Doc 89: scores[89] $= 1.3$

Step 2: Process "learning"

- Doc 5: scores[5] $= 1.5 + 0.8 = 2.3$
- Doc 27: scores[27] $= 1.6 + 1.5 = 3.1$
- Doc 35: scores[35] $= 0.9$

DAAT vs. TAAT Comparison

Aspect	DAAT	TAAT
Memory	Low (only top-k heap)	High (all candidate scores)
Cache locality	Good (sequential per doc)	Poor (random access)
Early termination	Easy (threshold pruning)	Hard (need all terms)
Parallelization	Harder	Easier (independent terms)
Skip pointers	Very effective	Less effective
Long queries	Better	Worse (more memory)
Conjunctive queries	Natural	Must filter at end
Modern systems	**Preferred**	Used in batch scoring

Practical Choice

- **DAAT:** Most production systems (Lucene, Elasticsearch, Solr)
- **TAAT:** Batch re-ranking, research systems

Query Optimization: MaxScore

Observation

Not all query terms contribute equally to final scores

MaxScore Algorithm:

1. Compute maximum possible contribution for each term

$$\text{maxScore}(t) = \max_{d} \text{score}(t, d)$$

2. Sort query terms by maxScore (descending)
3. Maintain threshold θ = score of k-th best document so far
4. Skip processing term t for document d if:

$$\text{score}(d) + \text{maxScore}(t) < \theta$$

(Can't possibly make top-k)

Result: Avoid scoring low-impact terms for documents that won't reach top-k

MaxScore Example

Query: "neural network learning"

Max scores:

- neural: maxScore $= 2.5$
- network: maxScore $= 2.0$
- learning: maxScore $= 1.5$

Sorted by maxScore: [neural, network, learning]

Processing doc 42 (top-10 query):

1. Current 10th-best score: $\theta = 5.0$
2. Score so far for doc 42: 3.0 (from "neural" and "network")
3. Max additional score: 1.5 (from "learning")
4. Total possible: $3.0 + 1.5 = 4.5 < 5.0$
5. **Skip** scoring "learning" for doc 42 (can't reach top-10)

Query Optimization: WAND

Weak AND (WAND)

Advanced optimization combining DAAT with dynamic pruning

Key ideas:

1. Maintain upper bounds on term contributions
2. Process posting lists in parallel (like DAAT)
3. Skip entire documents if upper bound ¡ threshold
4. Dynamically update threshold as better docs found

WAND pivot algorithm:

- Find "pivot" document where sum of upper bounds $\geq \theta$
- All documents before pivot can be safely skipped
- Use skip pointers to jump to pivot

Impact-Ordered Indexes

Alternative Index Organization

Instead of sorting postings by docID, sort by impact (score)

Traditional (docID-ordered):

- neural: [(5, tf=1), (12, tf=3), (27, tf=1), (89, tf=5)]
- Sorted by docID

Impact-ordered:

- neural: [(89, tf=5), (12, tf=3), (5, tf=1), (27, tf=1)]
- Sorted by term frequency (or TF-IDF score)

Benefits:

- **Early termination:** Stop after processing top-k high-impact postings

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Why Compress Indexes?

Motivation

1. **Storage costs**
   - Uncompressed index: 10-20% of raw collection size
   - For 100TB collection: 10-20TB index
   - Compression: 5-10× reduction → 1-4TB
2. **I/O efficiency**
   - Disk/SSD bandwidth: 500 MB/s
   - Decompression: 2-5 GB/s
   - Reading compressed data is **faster**!
3. **Cache efficiency**
   - More index fits in RAM
   - Better cache hit rates
   - Faster query processing

What to Compress?

Dictionary

Size: 100-500 MB (small)
Techniques:

- Front coding
- Suffix arrays
- Finite-state automata

Impact: Minor (already small)

Postings Lists

Size: Gigabytes to terabytes
Techniques:

- Gap encoding
- Variable-byte
- Gamma/Delta codes
- PForDelta

Impact: **Major** (most of index)

Typical Index Breakdown

- Dictionary: 2%
- Postings (docIDs): 70%

Dictionary Compression: Front Coding

Observation

Sorted dictionary has many shared prefixes

Uncompressed:

```
algorithm
algorithmic
algorithmically
algorithms
alien
align
alignment
```

Storage: 64 bytes

Front coded:

```
9 algorithm
  2 ic
  5 ally
  1 s
5 ali
  2 en
  2 gn
  5 ment
```

Storage: $\approx$ 38 bytes

Postings Compression: Gap Encoding

Key Insight

DocIDs in postings are sorted $\rightarrow$ store gaps instead of absolute values

Original postings (docIDs):

$$[5, 127, 214, 215, 512, 1024, 2048, 10239]$$

Gaps (d-gaps):

$$[5, 122, 87, 1, 297, 512, 1024, 8191]$$

Why This Helps

- **Original:** Numbers range from 5 to 10,239 (need 14 bits each)
- **Gaps:** Most numbers small (need fewer bits)
- Small numbers compress much better!

Gap Distribution

Zipf's Law

Term frequencies follow power-law distribution

Implications for gaps:

- **Rare terms:** Large gaps (appear infrequently)
- **Common terms:** Small gaps (appear often)
- **Overall:** Most gaps are small!

Example

Rare term "syzygy": [15, 5023, 89234]
Gaps: [15, 5008, 84211] (large gaps)

Common term "the": [1, 2, 5, 7, 8, 10, 12, ...]
Gaps: [1, 1, 3, 2, 1, 2, 2, ...] (small gaps)

Variable Byte (VB) Encoding

Core Idea

Use variable number of bytes to encode integers

- Small numbers: 1 byte
- Larger numbers: 2+ bytes
- Self-delimiting: continuation bit indicates if more bytes follow

Encoding scheme:

- Use 7 bits per byte for data
- Use 1 bit (MSB) as continuation flag
- 1 = last byte, 0 = more bytes follow

Number	Binary	VB Encoding
5	101	1_0000101 (1 byte)

VB Encoding: Detailed Examples

Example 1: Encode 5

1. Binary: 101 (3 bits)
2. Fits in 7 bits → use 1 byte
3. Data: 0000101, continuation: 1 (last byte)
4. Result: 10000101

Example 2: Encode 824

1. Binary: 1100111000 (10 bits)
2. Needs 2 bytes (10 ¿ 7)
3. Split into 7-bit chunks (from right): 0111000, 0000110
4. First byte (higher bits): 0_0000110 (more bytes follow)
5. Second byte (lower bits): 1_0111000 (last byte)
6. Result: 00000110 10111000

Decoding:

VB: Compressing a Posting List

Original docIDs: [5, 127, 214, 215]

Step 1: Compute gaps

- Gap 1: 5 - 0 = 5
- Gap 2: 127 - 5 = 122
- Gap 3: 214 - 127 = 87
- Gap 4: 215 - 214 = 1

Gaps: [5, 122, 87, 1]

Step 2: VB encode each gap

- 5 → 10000101 (1 byte)
- 122 → 11111010 (1 byte)
- 87 → 11010111 (1 byte)
- 1 → 10000001 (1 byte)

Result: 4 bytes compressed vs. 16 bytes uncompressed (32-bit ints)

Gamma Encoding

Motivation

VB uses 8 bits to encode 7 bits (12.5% overhead)
Can we do better for very small numbers?

Gamma encoding:

1. Let $N = \lfloor \log_2 x \rfloor$ (position of highest bit)
2. Encode N in unary: N zeros + one 1
3. Encode offset $x - 2^N$ in N bits

Examples

Encode 5:

- Binary: 101, $N = 2$ (highest bit at position 2)
- Unary for $N = 2$: 001 (two zeros + 1)

Gamma Encoding: More Examples

x	**Binary**	N	**Offset**	**Gamma Code**
1	1	0	0	1 (1 bit)
2	10	1	0	010 (3 bits)
3	11	1	1	011 (3 bits)
4	100	2	0	00100 (5 bits)
5	101	2	1	00101 (5 bits)
9	1001	3	1	0001001 (7 bits)
100	1100100	6	36	000000_1100100 (13 bits)

Observations:

- Very efficient for small numbers (1-15)
- Encoding length: $2\lfloor \log_2 x \rfloor + 1$ bits
- Self-delimiting (no need for length prefix)
- Optimal for geometric distributions

VB vs. Gamma Comparison

Number	VB (bits)	Gamma (bits)	Better
1	8	1	Gamma
5	8	5	Gamma
10	8	7	Gamma
100	8	13	VB
127	8	13	VB
128	16	15	Gamma
1000	16	19	VB
10000	16	27	VB

Trade-offs

- **Gamma:** Better compression for numbers ¡ 128
- **VB:** Simpler, faster, better for larger numbers
- **Gamma:** Requires bit-level operations (slower)

PForDelta: Patched Frame of Reference

Key Observation

In most posting lists, the vast majority of gaps fit in a small number of bits, but a few outliers are much larger

PForDelta idea:

1. Divide postings into blocks (e.g., 128 values)
2. Choose bit width b that fits 90% of values in block
3. Store majority in b bits each (packed tightly)
4. Store exceptions separately with positions

Example

Gaps: [5, 3, 7, 512, 2, 4, 1, 6, 9, 3]

- Choose $b = 3$ bits (fits all except 512)

PForDelta: Detailed Algorithm

Encoding:

```
Divide gaps into blocks of size B (e.g., 128)
for each block do
    Compute histogram of gap values
    Choose b such that ≥ 90% of values fit in b bits
    Pack values that fit into b-bit array
    For exceptions (values ≥ 2^b):
        Store position in block
        Store actual value separately
        Put special marker in main array
    Store: b, packed array, exception list
end for
```

Decoding:

1. Read b (bit width)

PForDelta: Performance

Advantages

- **Excellent compression:** 5-10× typical
- **Very fast decompression:** SIMD-friendly (process 128 bits at once)
- **Random access:** Can decode individual blocks
- **Adaptive:** Bit width chosen per block

Disadvantages

- **Complex implementation:** Bit packing, exception handling
- **Block-based:** Small overhead for block headers
- **Not universal:** Requires careful tuning

Typical Performance

- Compression: 6-8× for typical web collections

Compression Method Summary

Method	Compression	Dec. Speed	Complexity	Use Case
None	1×	Fastest	Trivial	Small indexes
VB	3-4×	Fast	Simple	General purpose
Gamma	4-5×	Medium	Moderate	Small gaps
PForDelta	6-10×	Very Fast	Complex	Production
Simple-9/16	4-6×	Very Fast	Moderate	Production

Recommendations

- **Learning/prototyping:** Start with VB (easy to implement)
- **Production (small/medium):** VB (good enough, simple)
- **Production (large scale):** PForDelta or Simple-16 (best performance)
- **Research:** Experiment with adaptive schemes

From Sparse to Dense Retrieval

Sparse (Traditional)

Representation:

- Bag of words
- High-dimensional (vocab size)
- Mostly zeros (99%+ sparse)

Index: Inverted index
Search: AND/OR logic, BM25
Limitation: Vocabulary mismatch

Dense (Neural)

Representation:

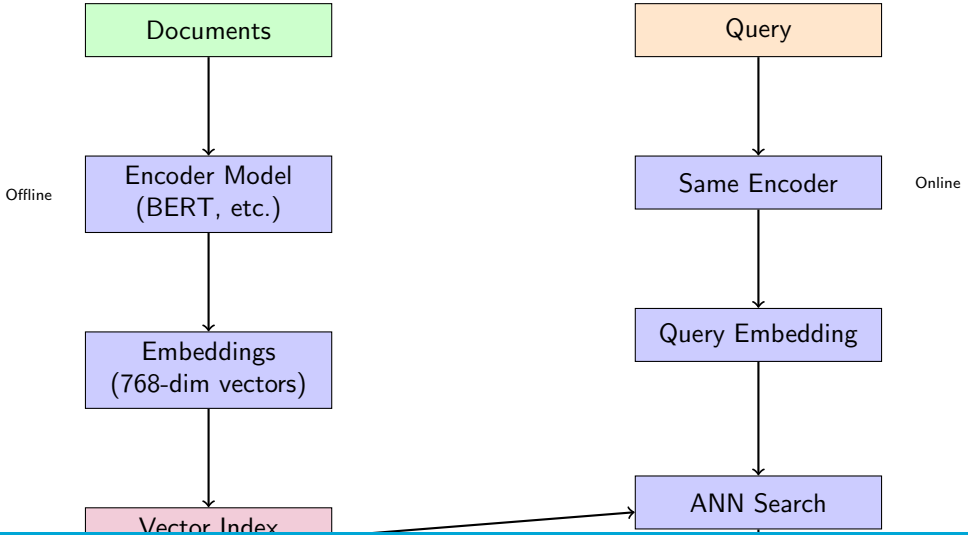
- Learned embeddings
- Low-dimensional (100-1000d)
- All non-zero (dense)

Index: Vector database
Search: Nearest neighbors
Advantage: Semantic matching

Example

Query: "car problems"
Sparse: Misses document with "automobile issues"
Dense: Finds it! (similar embeddings)

Dense Retrieval: The Pipeline



The k-NN Search Problem

Problem Definition

Given:

- Collection of N document vectors: $\{\vec{d}_1, \vec{d}_2, \ldots, \vec{d}_N\}$
- Query vector: $\vec{q}$
- Distance/similarity function: $\text{sim}(\vec{q}, \vec{d})$ (e.g., cosine, dot product)

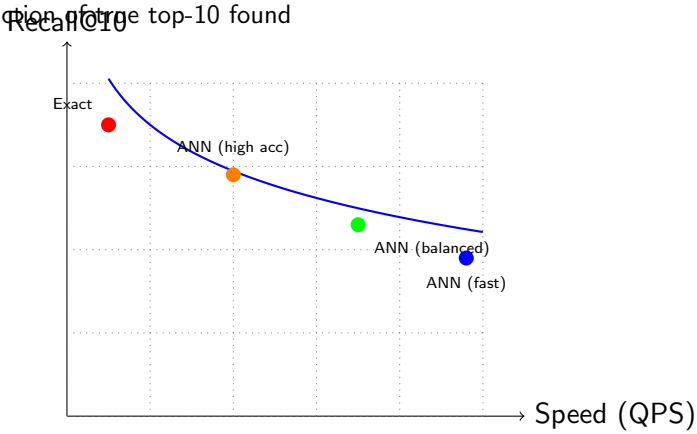
Find: k documents with highest similarity to $\vec{q}$

The Challenge

Exact search: Compare query to all N vectors $\rightarrow O(N \cdot d)$ where $d =$ dimension
For $N = 10^9$ docs, $d = 768$:
$\rightarrow$ 768 billion operations per query
$\rightarrow$ Too slow! (¿1 second even on GPU)

ANN: The Accuracy-Speed Trade-off

Recall@10: fraction of true top-10 found



More approximate → faster but less accurate

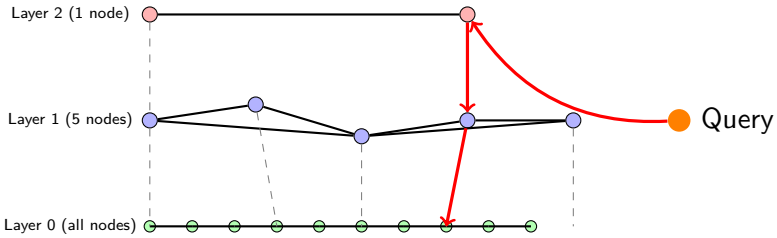
Main ANN Approaches

1. **Graph-Based** (HNSW, NSG)
   - Build graph connecting similar vectors
   - Search by navigating graph
   - High accuracy, moderate memory
2. **Clustering/Inverted File (IVF, IMI)**
   - Cluster vectors (k-means)
   - Search only in nearest clusters
   - Fast, lower accuracy
3. **Product Quantization (PQ)**
   - Compress vectors aggressively
   - Approximate distances in compressed space
   - Very memory-efficient
4. **Tree-Based** (Annoy, KD-tree)
   - Partition space with trees
   - Search by tree traversal
   - Simple, good for moderate dimensions
5. **Hashing** (LSH)

HNSW: Hierarchical Navigable Small World

Core Idea

Build a multi-layer graph where upper layers provide "highways" for fast navigation to the target region



Search: Start at top, greedily navigate to nearest neighbor at each layer, descend to next layer, repeat

HNSW: Construction

```
Input: Set of vectors V, parameter M (max connections per node)
Output: Multi-layer graph

Initialize empty graph
for each vector v⃗ in V do
    ℓ ← random layer (exponentially decaying probability)
    Insert v⃗ at layer ℓ and all layers below
    for each layer from top to ℓ do
        Find M nearest neighbors at this layer (using greedy search)
        Connect v⃗ to these neighbors
        if neighbor has ¿ M connections then
            Prune least useful edge
        end if
    end for
end for
```

HNSW: Search

```
Input: Query q⃗, graph G, k (neighbors to find), ef (search width)
Output: Approximate k nearest neighbors

currLayer ← top layer
entryPoint ← entry node at top layer
candidates ← priority queue (min-heap by distance)
visited ← set of visited nodes
while currLayer ≥ 0 do
    Add entryPoint to candidates and visited
    while candidates not empty do
        nearest ← pop closest unvisited from candidates
        for each neighbor n of nearest at currLayer do
            if n not in visited then
                Add n to candidates and visited
                if distance(q⃗, n) ; distance to farthest in top-ef then
```

HNSW: Properties

Advantages

- ✓ Excellent recall (95-99% typical)
- ✓ Fast search: $O(\log N)$ average
- ✓ Supports updates (add/delete vectors)
- ✓ No training required
- ✓ Works well in high dimensions (100-1000d)

Disadvantages

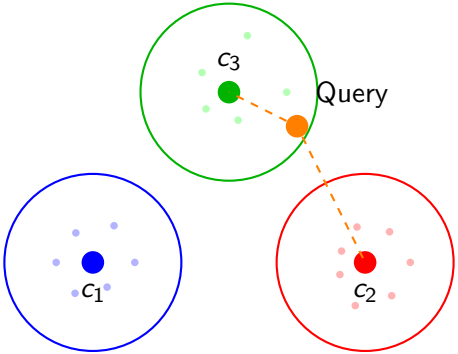
- ✗ High memory usage (graph structure: $\approx 2\times$ vector size)
- ✗ Slower to build than some alternatives
- ✗ Not as memory-efficient as quantization methods

When to use:

IVF: Inverted File Index

Core Idea

Cluster vectors with k-means, then search only in nearest clusters



Search only in 1-2 nearest clusters

IVF: Algorithm Details

Index Construction:

```
Run k-means on all vectors → get k cluster centers {c_1, ..., c_k}
Assign each vector to its nearest cluster center
Build inverted lists: cluster i → list of vectors in that cluster
```

Search:

```
Input: Query q⃗, n_probe (clusters to search), k (neighbors)
Compute distance from q⃗ to all cluster centers
Select n_probe nearest cluster centers
candidates ← []
for each selected cluster do
    Fetch all vectors in cluster
    Compute exact distances to q⃗
    Add to candidates
end for
Sort candidates by distance
return top-k candidates
```

Key parameter: n_{probe} (trade recall for speed)

- $n_{probe} = 1$: Fast but low recall (search 1 cluster)
- $n_{probe} = 10$: Balanced (search 10 clusters)
- $n_{probe} = k$: Exact search (search all clusters)

Product Quantization (PQ)

Goal

Compress vectors aggressively while preserving distance relationships

Key idea:

1. Split each d-dimensional vector into m subvectors (d/m dims each)
2. Quantize each subvector independently using k-means
3. Represent vector as m cluster IDs (e.g., m bytes)

Example

Original: 768-dim vector (768×4 bytes $= 3072$ bytes)
PQ compression:

- Split into $m = 96$ subvectors of 8 dims each
- Quantize each subvector to $k = 256$ centroids (1 byte ID)
- Compressed: 96 bytes ($32\times$ compression!)

Distance computation:

- Pre-compute distances from query subvectors to all centroids
- Approximate distance = sum of subvector distances (lookup table)
- Very fast! (just m additions)

PQ: Detailed Algorithm

Training:

```
Split each vector into m subvectors of dimension d/m
for each subspace i = 1, ..., m do
    Run k-means on all subvectors in subspace i → get k centroids
    Store these centroids as codebook C_i
end for
```

Encoding:

```
Input: Vector v⃗
Split v⃗ into subvectors [v⃗_1, v⃗_2, ..., v⃗_m]
for each subvector v⃗_i do
    Find nearest centroid in C_i → get ID c_i
end for
Store vector as [c_1, c_2, ..., c_m] (m bytes if k=256)
```

IVF + PQ: Best of Both Worlds

IVFPQ: Combine IVF with Product Quantization

1. Use IVF for coarse filtering (cluster-level search)
2. Use PQ for compression within clusters
3. Get both speed (IVF) and memory efficiency (PQ)

Algorithm:

1. Offline: Cluster vectors into k clusters (IVF)
2. Offline: Within each cluster, train PQ codebooks
3. Offline: Encode all vectors with PQ
4. Online: Find n_{probe} nearest clusters
5. Online: Search within those clusters using PQ distances

Memory savings example:

- Original: 1B vectors $\times$ 768 dims $\times$ 4 bytes $=$ 3TB
- IVFPQ: 1B vectors $\times$ 64 bytes (PQ) $=$ 64GB
- **50$\times$ compression!**

Trade-off: Lower recall than HNSW, but much more memory-efficient

ANN Libraries Comparison

Library	Method	Recall	Speed	Memory	Updates
HNSW (hnswlib)	Graph	Excellent	Fast	High	Yes
FAISS (IVF)	Clustering	Good	Very Fast	Medium	Limited
FAISS (IVFPQ)	Clustering+PQ	Good	Very Fast	Low	Limited
FAISS (HNSW)	Graph	Excellent	Fast	High	Yes
Annoy	Trees	Good	Fast	Medium	No
ScaNN	Learned quant.	Excellent	Very Fast	Low	Limited
Milvus	Multiple	Excellent	Fast	Varies	Yes
Weaviate	HNSW	Excellent	Fast	High	Yes

Choosing a Library

- **High recall needed:** HNSW (hnswlib, Weaviate)
- **Memory constrained:** FAISS IVFPQ, ScaNN
- **Maximum speed:** FAISS IVF (lower recall)
- **Distributed/production:** Milvus, Weaviate, Qdrant
- **Simple/prototyping:** Annoy, hnswlib

Hybrid: Sparse + Dense

Why Hybrid?

Sparse and dense retrieval have complementary strengths

Sparse (BM25) strengths:

- Exact term matching
- Rare/specific terms
- Named entities
- Acronyms, IDs
- Explainable

Dense strengths:

- Semantic matching
- Synonyms
- Paraphrases
- Conceptual similarity
- Cross-lingual

Hybrid approaches:

1. **Parallel retrieval + fusion**
   - Run sparse and dense in parallel
   - Merge results (e.g., reciprocal rank fusion)

Key Takeaways: Posting Lists

1. **Posting list structure**
   - Core: (docID, frequency, positions)
   - Sorted by docID for merge operations
   - Variable length (power-law distribution)
2. **Skip lists**
   - Add "highways" for faster intersection
   - Place every $\sqrt{n}$ elements typically
   - Trade space for speed

Key Takeaways: Query Processing

1. **DAAT vs. TAAT**
   - DAAT: Process one document at a time (modern systems)
   - TAAT: Process one term at a time (batch scoring)
2. **Optimizations**
   - MaxScore: Skip low-impact terms
   - WAND: Dynamic pruning with upper bounds
   - Impact-ordered indexes: Early termination

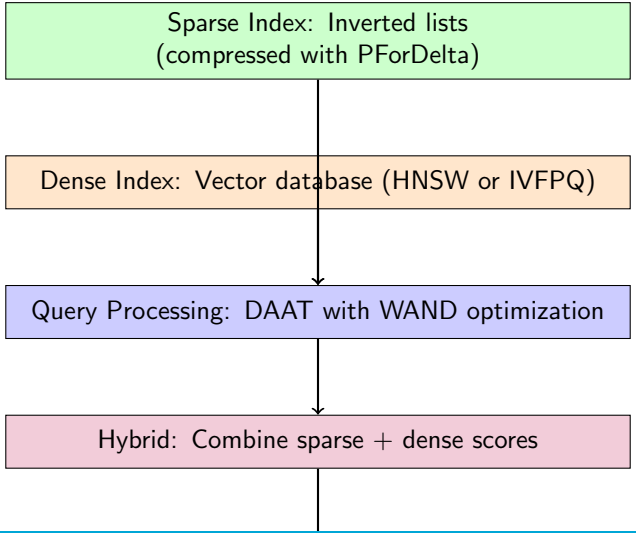
Key Takeaways: Compression

1. **Gap encoding is essential**
   - Store gaps instead of absolute docIDs
   - Most gaps are small (compress well)
2. **Compression methods**
   - VB: Simple, fast, 3-4$\times$ compression
   - Gamma: Better for small numbers
   - PForDelta: Best performance, 6-10$\times$ compression
3. **Why it matters**
   - Smaller index $\rightarrow$ fits in RAM
   - Faster I/O $\rightarrow$ better query latency
   - Disk is slow, CPU is fast

Key Takeaways: ANN Indexing

1. **Dense retrieval requires ANN**
   - Exact search too slow for billion-scale
   - Trade recall for speed (90-99% recall acceptable)
2. **Main approaches**
   - HNSW: Graph-based, excellent recall
   - IVF: Clustering, fast but needs more probes
   - PQ: Aggressive compression, memory-efficient
   - IVFPQ: Combines IVF + PQ (production standard)
3. **Hybrid is often best**
   - Sparse for exact matching
   - Dense for semantic matching
   - Combine for best of both worlds

The Complete Modern IR Stack



Questions?

Thank you!

Questions?

Next: Lecture 3 — Ranking Models

We'll use these indexes to implement BM25, language models, and query expansion with RM3!